



Unveiling M-A chain connectivity: key to optimizing low-temperature toughness in narrow-gap laser wire-feed welding offshore steel

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ABSTRACT

Narrow-gap laser wire-feed welding shows promise for welding thick offshore steel plates, but the origins of heat-affected zone (HAZ) toughness remain controversial. This study examined 20 mm thick EH36 steel joints made under different laser power conditions. Combined characterization techniques were used to explore the relationships between power and toughness and their underlying mechanisms. The results show that the impact toughness changed nonmonotonically with increasing laser power, reaching a peak at approximately 5.5 kW, whereas a shift from dimple to quasicleavage fracture occurred at 6.5 kW, causing the toughness to decrease. The main reason for the toughness loss was identified as the chain-like martensite-austenite constituents in the intercritical HAZ (ICHAZ). Clarifying the chaining coefficient and conducting quantitative analysis on the average chain length, combined with spatial correlation to high-KAM bands, confirmed that failure occurred mainly along interfaces: the elastoplastic mismatch between the martensite-austenite constituent (M-A) and ferrite increased interfacial stress triaxiality, leading to the formation of voids preferentially at the ends of lath-like M-A and chain intersections. These then propagated quickly along the M-A chains and coalesced. Therefore, the enhanced crack connectivity through the M-A chains in the ICHAZ, rather than the grain size in the coarse-grained HAZ (CGHAZ), mostly explained the decrease in toughness at high power.

1. Introduction

As marine engineering equipment moves toward deeper water and larger scales globally, the low-temperature toughness and service reliability of welded joints in offshore steel have become crucial factors limiting equipment performance. Offshore steels, such as EH36 steels, which are known for their excellent strength and toughness balance, are widely used in critical load-bearing structures such as ship hulls and deep-sea oil and gas platforms [1]. However, welding plates thicker than 20 mm present significant challenges. Traditional arc welding methods such as submerged arc welding and TIG welding with a high heat input can cause HAZ grain coarsening and the segregation of martensite-austenite (M-A) constituents. This results in a notable deterioration in the low-temperature impact toughness [2–3]. Furthermore, traditional welding techniques tend to result in uneven energy distributions, producing shallow yet broad welds that compromise both the strength and sealing integrity [4–5].

Compared with traditional welding methods, laser welding

technology addresses these limitations. Its benefits, such as high energy density, low heat input, and a narrowed HAZ (width ≤ 2 mm), make it a key area of research for efficient, low-damage welding of thick plates [6]. For example, Katayama et al. [7] achieved single-pass welding of 70 mm thick stainless steel via a high-power laser in a low-vacuum environment, but the process parameters (such as extremely high laser power and very low welding speed) make it difficult to meet industrial production needs. Lei et al. further advanced dual-beam laser welding technology, doubling the welding efficiency for 120 mm S960 high-strength steel [8]. Therefore, narrow-gap laser wire feeding welding (NGLWFW) has distinct advantages over conventional laser welding processes in thick-plate welding because it combines multipass filling techniques with precise heat input control. Hou et al. [9] successfully welded 100 mm ultrathick offshore steel via the NGLWFW process, and the joint met impact toughness requirements even at -80 °C. However, these studies focused mainly on optimizing process parameters and characterizing macroproperties and lacked a systematic analysis of microstructural gradient changes in the HAZ during multipass welding

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and its relationship with low-temperature toughness.

Research on multilayer, multipass welding indicates that the HAZ, which undergoes multiple thermal cycles, experiences performance degradation due to grain coarsening, M-A constituent enrichment, and uneven microstructure distribution. This makes it the weakest part of the welded joint. For example, Zhu et al. [10] reported that upper bainite and martensite primarily enhance strength, whereas granular bainite and acicular ferrite improve toughness by hindering crack growth. Jia et al. [11] studied the microstructure and toughness of different HAZ subzones and revealed that the shape and distribution of M-A constituents and the density of high-angle grain boundaries are key factors affecting HAZ performance. Arora et al. [12] reported that secondary thermal cycles in the intercritical region can increase the volume fraction of grain boundary M-A constituents to 15–20%, reducing the impact toughness by more than 50%. However, most traditional studies are based on single-pass welding or thermal simulation experiments, which do not adequately reflect the cumulative effect of gradient heat input on the HAZ microstructure in thick plate multipass welding. Wang et al. [13] demonstrated that the necklace-like M-A constituents formed in the reheated zone of multipass welds are the main brittle areas, with their size, distribution, and volume fraction directly affecting toughness. Multiple thermal cycles can partially decompose M-A, thereby improving the impact toughness. Additionally, the rapid cooling characteristic of laser welding quickly induces martensite hardening and brittle phase formation, but current research lacks a detailed exploration of the grain size gradient, M-A constituent morphology, and solute element enrichment under NGLFW conditions [14,15]. To gain a deeper understanding of damage induced by hard phases, recent advanced characterisations have provided critical mechanistic insights. Zhang et al. [16] employed in situ EBSD and DIC techniques to reveal that strain incompatibility between hard M-A islands and the soft matrix leads to severe lattice rotation and dislocation pile-up, directly triggering boundary decohesion. Du et al. [17] further confirmed through in situ SEM observations that this incompatible deformation between hard particles and the ductile matrix is the main cause of local stress concentration and crack initiation. Giang et al. [18] quantitatively demonstrated through numerical simulations that fracture risk is not solely governed by particle fraction, but is also critically amplified by the distribution of brittle phases, which effectively increases microstructural damage size. However, despite these micromechanical insights, current research on NGLFW remains limited to characterizing the individual morphology or volume fraction of M-A constituents. Crucially, the 'spatial connectivity' of these brittle phases has been largely overlooked. Consequently, a quantitative understanding of how these interconnected microstructures guide crack propagation in the HAZ is still lacking, limiting the development of effective strategies to suppress brittle fracture in thick-plate offshore steels.

Therefore, this work hypothesizes that the spatial connectivity of M-A constituents, rather than merely their volume fraction or individual morphology, governs the crack propagation path and thus the low-temperature toughness in the ICHAZ of multipass NGLFW joints. To verify this, the present study utilizes 20 mm thick EH36 steel to design NGLFW experiments with varying heat inputs. By employing multi-scale characterization techniques such as optical microscopy (OM), scanning electron microscopy (SEM), and electron backscatter diffraction (EBSD), this study systematically quantifies the M-A chaining behavior. It also explores its relationship with strain localization and low-temperature impact toughness. This research provides new insights into the mechanisms that can aid in the design of efficient welding processes that minimize damage in thick offshore steel plates.

2. Experimental methods and materials

The base material used for the welding experiments was 20 mm thick EH36 offshore steel produced from Ansteel. The plate dimensions were 200 mm × 100 mm × 20 mm. To ensure that the weld joint properties

were as consistent as possible with those of the base metal, an ER55-Ni3 filler wire was selected. It has a diameter of 1.2 mm and a melting point of 1493 °C. This type of wire offers high plasticity, toughness, and impact resistance, as well as excellent low-temperature impact toughness. The chemical compositions of the base metal and filler wire are listed in Table 1.

The welding experiments were conducted via an automated laser welding system produced by KUKA Automation GmbH in Germany. The system is equipped with a fully automatic laser control device and a modular laser unit that significantly improves weld quality. The system consists of submodules, including the laser unit, wire feeder, clamping device, and travel mechanism. The laser source was a YLS-30000 multimode fibre laser from Germany with a maximum power output of 10 kW. The wire feed speed was precisely controlled between 0.2 and 22 m/min. The wire was fed through a conduit to a welding torch mounted on a KUKA robotic arm. The travel control system is part of the KUKA robot system and features a six-axis welding robot with flexible movement and a positioning accuracy of 0.05 mm.

Through finite element simulation and preliminary experimental fitting, the process window was determined to be 3.5–6.5 kW on the basis of peak temperature and thermal cycling conditions. The welding process used was narrow-gap laser wire-feed welding. Prior to welding, the plates underwent preprocessing. A V-groove was machined at the butt joint, as shown in Fig. 1a. The plates were subsequently ground and cleaned to prevent impurities from affecting the quality of the weld. The welding procedure began with an autogenous root pass using a single 4 kW laser heat source. Then, different laser powers—ranging from 3.5 to 6.5 kW—were used to complete the filler passes within the groove. The welding speed (0.6 m/min), wire feed rate (6 m/min), and laser defocus distance (+10 mm) were held constant to isolate the effect of the laser power. The corresponding linear heat input ($E = P/v$) ranges from 3.5 to 6.5 kJ/cm. More detailed process parameters are shown in Table 2. To ensure the stability of the thermal cycle and minimize the cumulative heat effect on the HAZ microstructure, the interpass temperature was strictly monitored and controlled below 150 °C via an infrared thermometer. A total of five layers were welded. After the first pass was completed, the remaining four passes involved narrow-gap laser wire-filling welding. Welding of the subsequent pass was initiated only after the weld bead had cooled to the target temperature. Furthermore, the detailed experimental setup, including the 10 kW fibre laser system, the wire feeding unit, and the clamping device, is shown in Fig. 1b.

The impact tests were performed at -40 °C in accordance with the GB/T 229-2020 standard, as shown in Fig. 1c. Testing was carried out via a PIT series Type D pendulum impact testing machine. To ensure accurate temperature control, the samples were immersed in a liquid nitrogen and alcohol bath for 15 min prior to testing to achieve thermal equilibrium throughout the sample volume. The fracture surfaces of the tested samples were subsequently examined via SEM to analyse the fracture mechanisms.

The samples used for microstructural analysis were sectioned from the welded joints and prepared according to standard metallographic procedures. The samples were mechanically ground with SiC papers of increasing fineness up to 2000 grit and subsequently polished with a diamond suspension (2.5 and 0.5 µm) to achieve a mirror finish. Different etching protocols were employed depending on the observation method. For optical microscopy, the polished samples were etched with LePera reagent for approximately 3 min to effectively reveal the phase constituents, particularly the M-A constituents. For scanning electron microscopy (SEM) analysis, the samples were etched with a 4% nital solution for 12 s. Microstructural observations were performed via a JVCTK-350EG optical microscope and a field emission scanning electron microscope (FE-SEM, JSM-7800F, JEOL, Japan). Detailed crystallographic features were characterized via electron backscatter diffraction (EBSD). The EBSD data were acquired via an Oxford Nordlys Max3 detector mounted on the FE-SEM system. The data collection was performed at an accelerating voltage of 20 kV and a probe current of 14

Table 1

Chemical composition (wt.%) of EH36 steel and welding wire.

Element	C	Si	Mn	Cu	Ni	Cr	Nb	V	Ti	P	S
EH36	0.06	0.2	1.44	0.25	0.3	0.1	0.03	0.04	0.011	0.015	0.003
welding wire	0.1	0.6	1.15	—	3.35	—	—	—	—	0.01	0.008

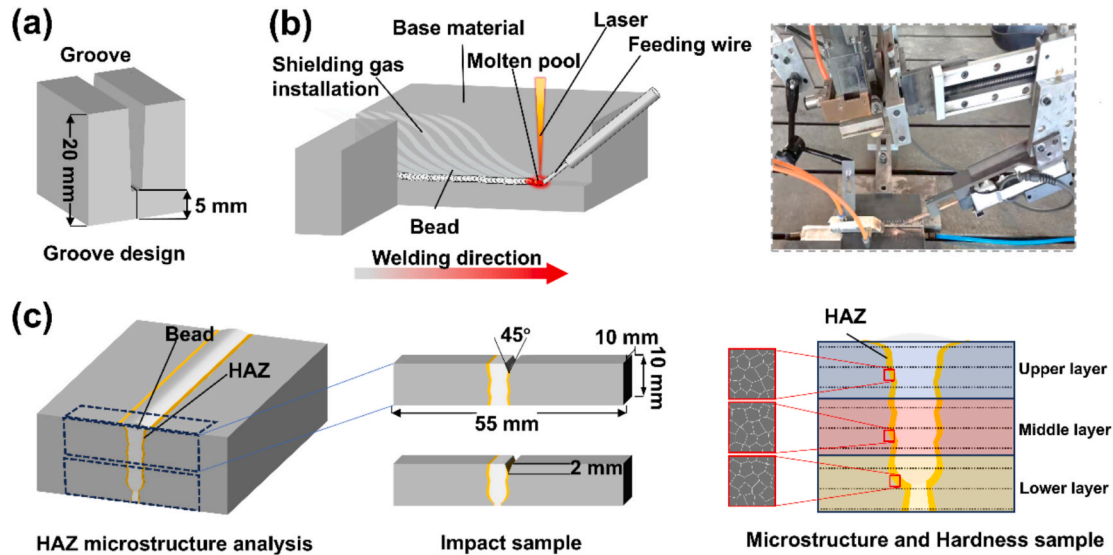


Fig. 1. (a) Cross-section of the groove design; (b) schematic of the LNGWFW unit and actual processing site; (c) impact test specimen sampling location, microstructure examination position and hardness testing.

Table 2

Detailed welding parameters for each sample.

Laser Power (kW)	Welding Speed (m/min)	Wire Feed Speed (m/min)	Defocus Distance (mm)	Linear Heat Input (kJ/cm)
3.5	0.6	6	+10	3.5
4.5	0.6	6	+10	4.5
5.5	0.6	6	+10	5.5
6.5	0.6	6	+10	6.5

nA. A step size of 0.5 μm was utilized to ensure sufficient resolution for calculating the effective grain size and kernel average misorientation (KAM). Postprocessing was conducted via AZtec software. For local strain assessment, the KAM was calculated using a maximum

misorientation threshold of 5° to exclude grain boundaries. Micro-hardness testing was conducted on various regions of welded joints prepared under different processing conditions. The testing procedure involved an HV0.5 load condition, with a loading duration of 10 seconds. Test points were distributed at 0.2 mm intervals perpendicular to the weld thickness. In the weld thickness direction, the spacing between adjacent test rows was set at 2 mm.

3. Results and discussion

3.1. Welding experiment

As shown in Fig. 2, the application of laser powers ranging from 3.5 kW to 4.5 kW results in the formation of macroscopic defects within

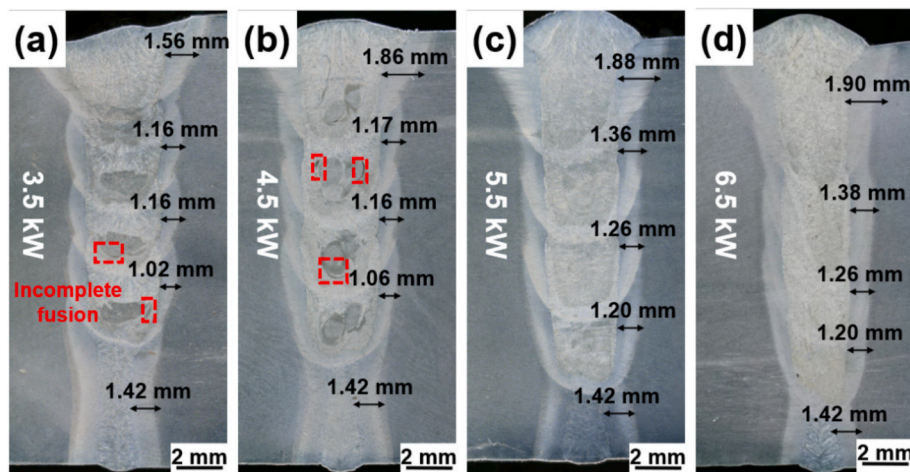


Fig. 2. Macrostructure of the weld cross section. (a) 3.5 kW laser power; (b) 4.5 kW laser power; (c) 5.5 kW laser power; (d) 6.5 kW laser power.

specific joint regions due to incomplete fusion, thereby impeding the precise analysis of the correlation between the microstructure and low-temperature impact toughness. A meticulous examination was conducted on the equipment, with no defects detected at either 5.5 kW or 6.5 kW. In future impact tests, samples fractured at the weld location will be excluded to obtain the reasonable low-temperature impact absorption energy of the heat-affected zone for 3.5 kW and 4.5 kW laser powers. This study focuses on 5.5 kW and 6.5 kW welding conditions, where the welds exhibit superior low-temperature impact toughness, more stable mechanical properties, and no defects. A comprehensive examination and comparative analysis of the microstructure, hardness and impact toughness of these welds were conducted.

3.2. Low-temperature impact tests

Fig. 3 shows the impact test findings for the HAZ using laser powers ranging from 3.5 to 6.5 kW. It should be specifically noted that the low-temperature impact absorption energy values for the 3.5 kW and 4.5 kW specimens shown in Fig. 3 were obtained from specimens confirmed to be free of macroscopic defects following fracture morphology inspection. Consequently, these results may be utilised to analyse the influence of laser power on the low-temperature toughness of the material. As the power increased from 3.5 to 5.5 kW, the impact toughness and absorbed energy increased from approximately 150 J to 240 J. At 6.5 kW, the toughness decreased to 176 J. This suggests that 5.5 kW is best for impact. At this power, the absorbed energy increased by more than 60% compared with 3.5 kW. Additionally, the difference in energy between the upper and lower HAZ regions decreased with increasing power. This shows that a small increase in the laser power can improve the uniformity of the mechanical properties in the HAZ.

To explore the nonmonotonic changes in the laser power and low-temperature toughness, the fracture surface morphology provides the first clue, as demonstrated in Fig. 4. As illustrated in Fig. 4a, the fracture surface in the upper region exhibited a 5.5 kW fracture energy and was characterized by deep, coherent dimples that were uniform in size and interconnected. This fracture surface exhibited typical ductile fracture characteristics. Conversely, at 6.5 kW, as illustrated in Fig. 4b, the fracture surface displayed distinct river patterns and cleavage steps, accompanied by localized intergranular brittle fracture. In the middle region, at 5.5 kW, as illustrated in Fig. 4c, the dimples became increasingly shallow, the tear ridges appeared blurred, and larger flat areas, referred to as plateaus, were observed. Despite the retention of certain plastic characteristics, a substantial reduction in toughness was observed. As demonstrated in Fig. 4d, at 6.5 kW, the presence of dimples was minimal, with cleavage platforms predominating. This observation is indicative of brittle cleavage fracture. In the lower region at 5.5 kW, as illustrated in Fig. 4e, the dimples displayed greater depth and size than those in the middle region. Concurrently, at 6.5 kW, as illustrated in Fig. 4f, cleavage-like steps became evident, suggesting a transition

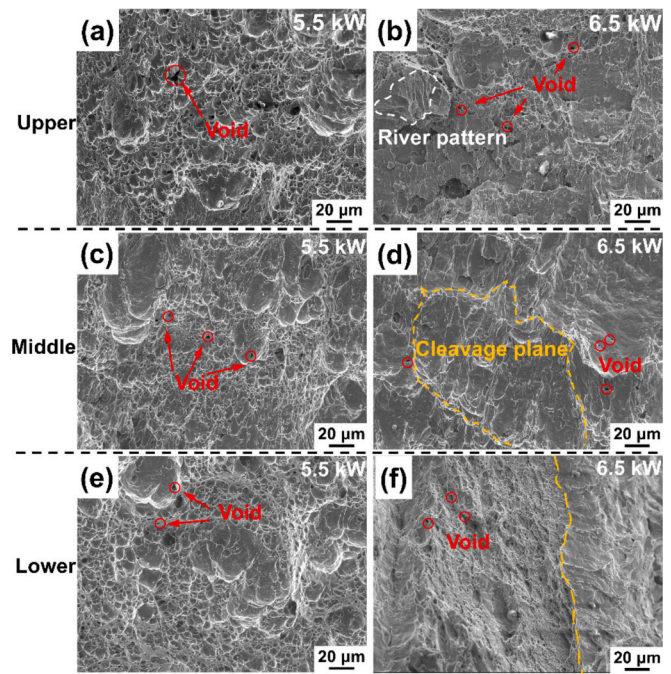


Fig. 4. SEM fractography of Charpy impact samples tested at -40°C . (a, c, e) Specimens welded at 5.5 kW taken from the upper, middle, and lower regions; (b, d, f) specimens welded at 6.5 kW taken from the upper, middle, and lower regions.

towards brittle fracture.

In general, as the laser power increased from 5.5 kW to 6.5 kW, the fracture mode transitioned from ductile to brittle across all regions. Furthermore, under conditions of identical power, the middle region consistently demonstrated the poorest low-temperature toughness compared with the upper and lower regions.

3.3. Hardness test

As shown in Fig. 5, the full-field Vickers microhardness distribution across the weld joint cross-section is presented at laser powers of 5.5 kW and 6.5 kW. The color gradient from deep blue to deep red serves to visually quantify the mechanical inhomogeneity within the narrow-gap laser-welded joint. As demonstrated in the illustration, both joints manifest a conspicuous sandwich-like hardness distribution pattern. The weld metal and adjacent CGHAZ—defined as the region within or near the white dashed fusion line—exhibit the highest hardness values, which are predominantly distributed between 260 and 300 HV_{0.5}. Conversely, a pronounced hardness gradient is evident, extending

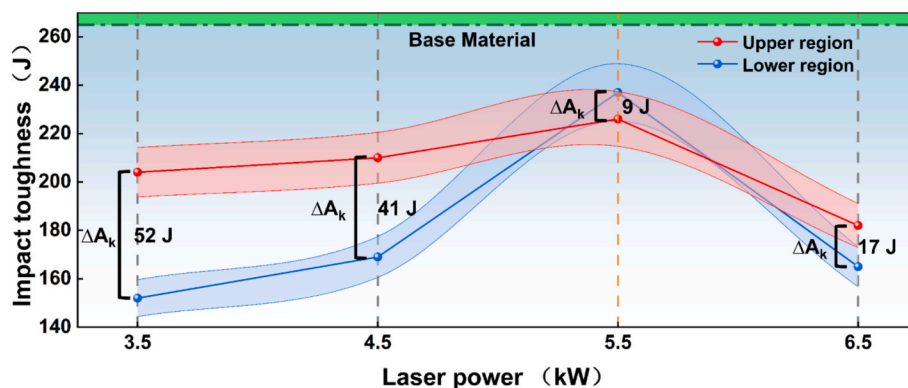


Fig. 3. Impact-absorbed energy in the HAZ at -40°C .

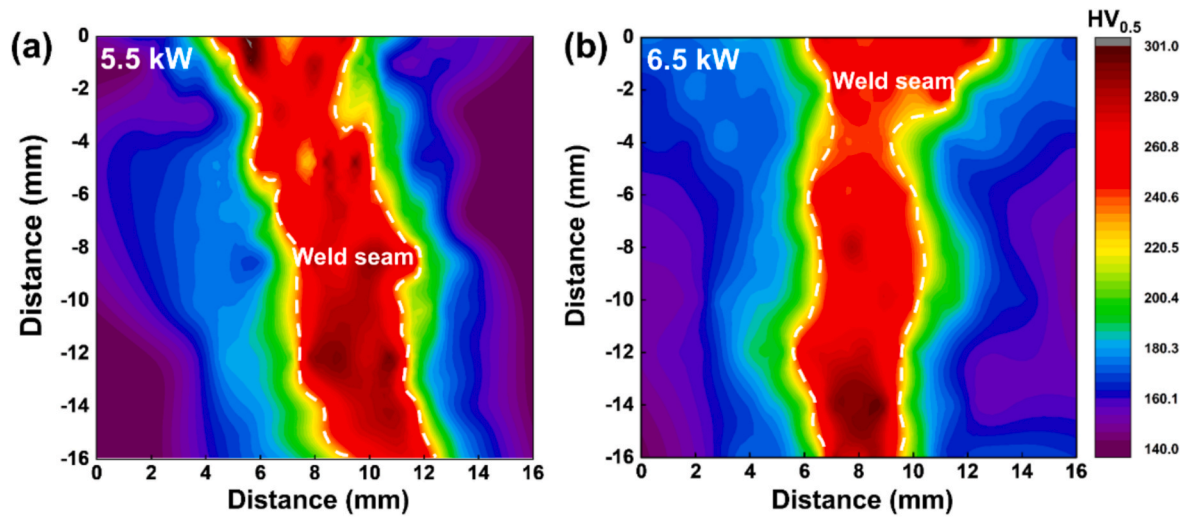


Fig. 5. Full-field cross-sectional Vickers microhardness mappings (HV0.5) of the narrow-gap laser wire-feed welded joints processed at different laser powers. (a) 5.5 kW; (b) 6.5 kW.

radially from the fusion line. The hardness of the material decreases sharply within the HAZ and ICHAZ and ultimately reaches the lowest baseline value at the base metal.

Upon comparing Figs. 5(a) and (b), the high-hardness zone in the joint with a heat input of 6.5 kW appears to be marginally wider. This is attributable to the elevated heat input, leading to an expansion of the

heat-affected zone. Notably, this characteristic distribution of the hardness gradient induces a macroscopic strength mismatch. In the context of impact loading, this nonuniformity leads to the localization of plastic strain within the relatively narrow and softer heat-affected zone.

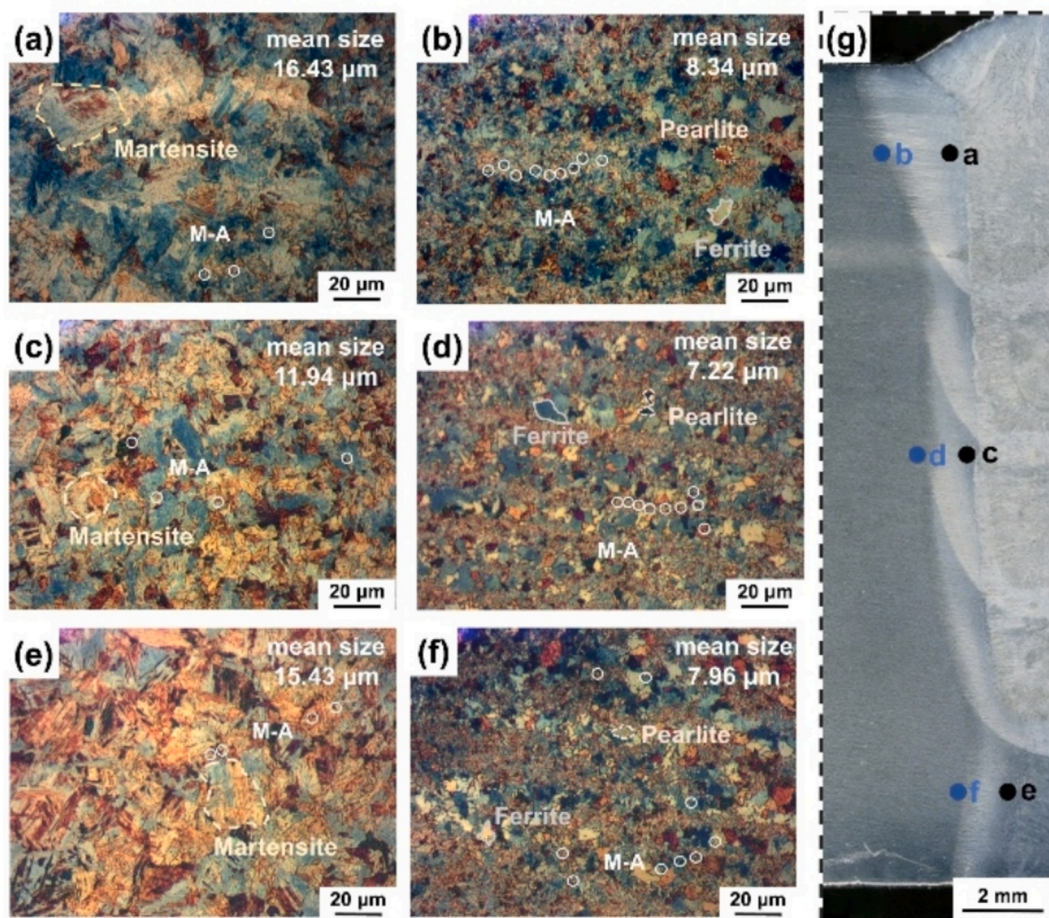


Fig. 6. Microstructural evolution in the upper CGHAZ and ICHAZ at a 5.5 kW laser power. (a) upper CGHAZ; (b) upper ICHAZ; (c) middle CGHAZ; (d) middle ICHAZ; (e) lower CGHAZ; (f) lower ICHAZ; (g) observation location.

3.4. Microstructure

Within the weld heat-affected zone, both the CGHAZ and the ICHAZ exhibit inferior mechanical properties compared with those of the fine-grained region. Consequently, under impact loading, fracture typically initiates within either the CGHAZ or ICHAZ. Figs. 6 and 7 show the microstructural characteristics of the upper, middle, and lower regions of the heat-affected zone at 5.5 kW and 6.5 kW of power. Under these process conditions, the microstructure of the CGHAZ in all regions consists primarily of martensite, accompanied by a small amount of granular bainite. This microstructure provides the high hardness of the coarse-grained heat-affected zone. The ICHAZ microstructure, however, comprises two components: retained coarse ferrite and newly formed fine microstructures. The new phases include ferrite, pearlite, and a small amount of martensite. This microstructural characteristic results in a hardness higher than that of the base material but with significantly greater ductility than the CGHAZ. However, both the ICHAZ and CGHAZ contain minute m-a components.

Critically, M-A constituents—which are detrimental to low-temperature toughness—have been observed in all regions [19]. However, their distribution patterns exhibited fundamental differences between the two zones. As shown in Fig. 8a-c, within the CGHAZ, although the M-A phase constituents were relatively large in size, they predominantly formed lamellar structures aligned along martensitic lamellae. This resulted in their relatively isolated distribution, confined within the martensitic grain boundaries. In stark contrast, as depicted in Fig. 8d-e, within the ICHAZ, the M-A phase components predominantly exhibit a fine-grained, block-like morphology. Nevertheless, they organize into a

continuous chain-like network along the ferrite grain boundaries.

3.5. Microstructure Beneath the Fracture Surface

As shown in Fig. 9, the microstructures of the sections beneath the impact fracture surface in the middle region are of particular interest. As shown in Fig. 9a, the crack path was tortuous at 5.5 kW, indicating superior low-temperature toughness. Conversely, at 6.5 kW, the crack path was relatively straight, suggesting reduced propagation resistance. In the crack initiation zone and adjacent areas, which are clearly visible in Fig. 9c and 9d, finely sized M-A constituents aggregated in chains and were distributed continuously along the grain boundaries, which is consistent with the distribution features of some M-A constituents in the ICHAZ. Furthermore, many voids, which formed during the process of fracture deformation, were observed to be present between the individual chain-aggregated M-A constituents that were found to surround the crack.

4. Discussion

4.1. Effect of M-A Constituents on the Low-temperature Toughness of the Weld HAZ

According to the classical Cottrell–Petch relationship[20], the refined microstructure of the ICHAZ, with a total grain boundary length of 29771 μm , should theoretically exhibit superior toughness compared with that of the CGHAZ (21,377 μm). However, the experimental results revealed that low-temperature impact cracks preferentially propagated

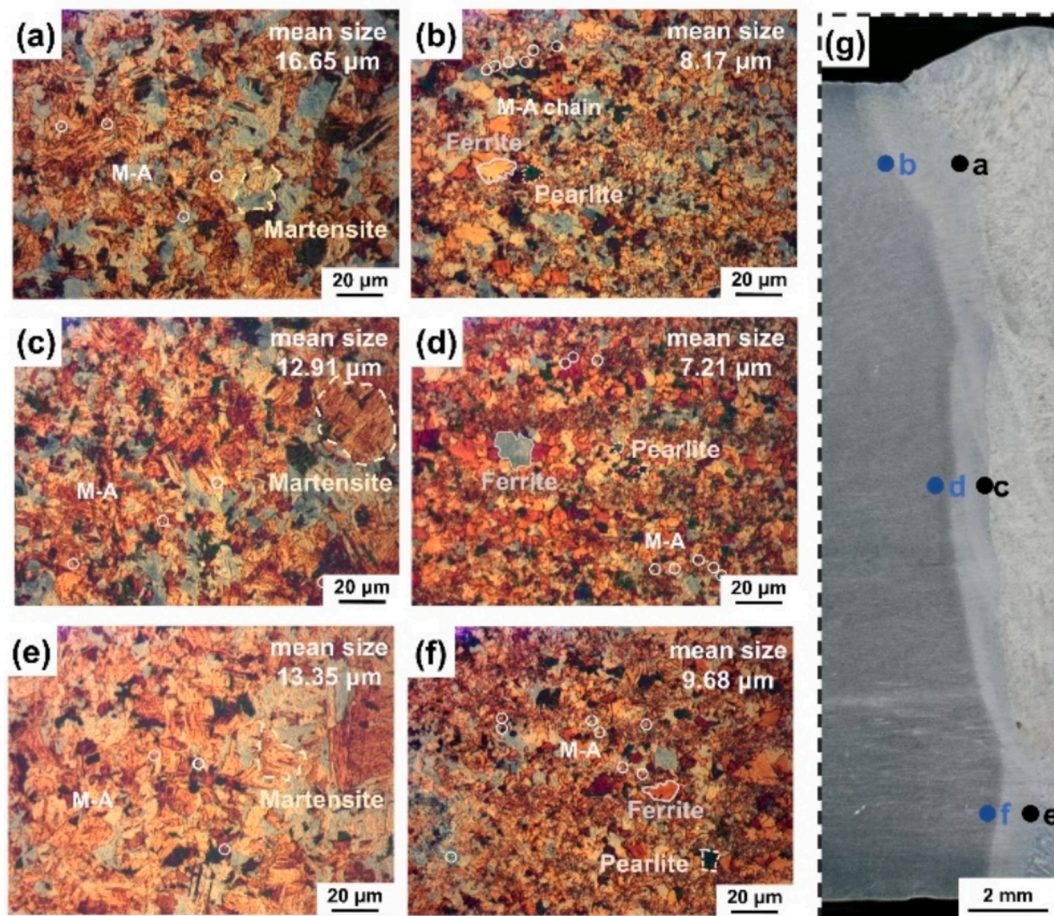


Fig. 7. Microstructural evolution in the upper CGHAZ and ICHAZ at a 6.5 kW laser power. (a) upper CGHAZ; (b) upper ICHAZ; (c) middle CGHAZ; (d) middle ICHAZ; (e) lower CGHAZ; (f) lower ICHAZ; (g) observation location.

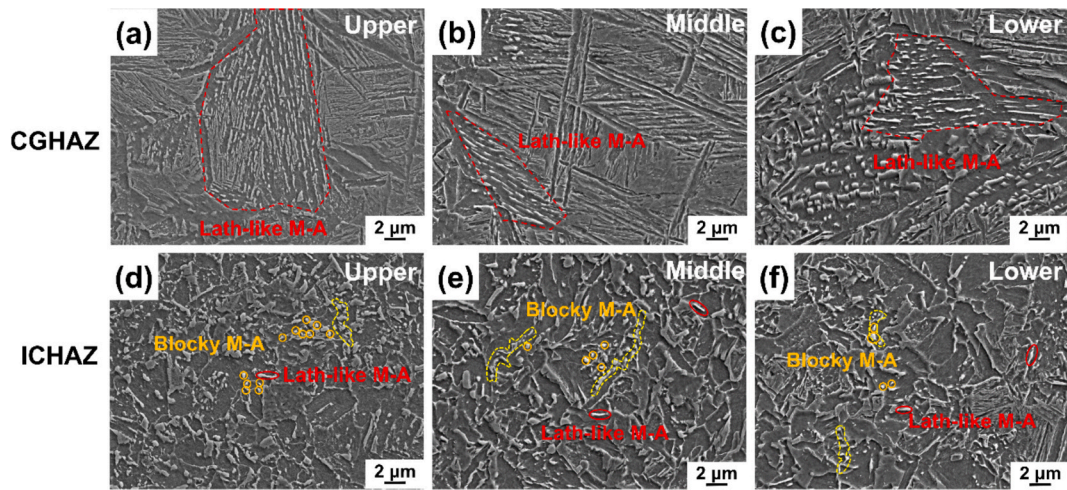


Fig. 8. SEM micrographs comparing the morphology and distribution of M-A constituents at 5.5 kW. (a) Upper CGHAZ; (b) middle CGHAZ; (c) lower CGHAZ; (d) upper ICHAZ; (e) middle ICHAZ; (f) lower ICHAZ.

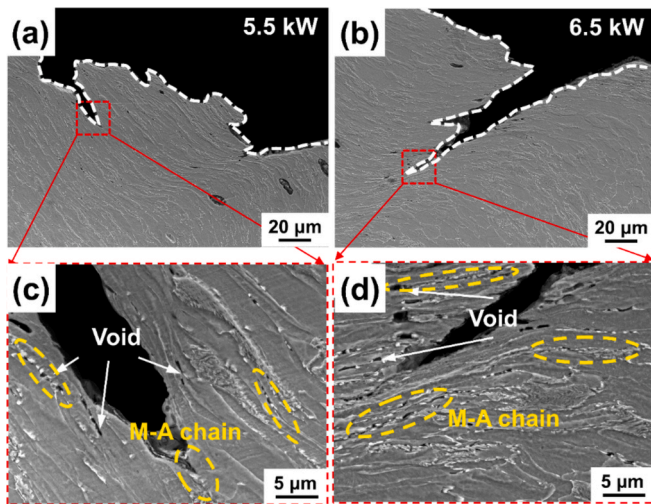


Fig. 9. Cross-sectional SEM micrographs of the subsurface microstructure immediately beneath the fracture surface tested at -40 °C. (a, c) 5.5 kW sample showing blunted secondary cracks arrested by the ferrite matrix; (b, d) 6.5 kW sample showing secondary cracks propagating preferentially along the M-A chains in the ICHAZ.

along the ICHAZ rather than the CGHAZ. This counterintuitive behavior suggests a transition in the governing toughening mechanism from being grain-size-dominated to being primarily controlled by phase interface connectivity and strain localization.

The EBSD analysis in Fig. 10 elucidates the crystallographic rationale for this failure. In the ICHAZ, the Schmid factor (SF) drastically fluctuates due to the coexistence of soft ferrite and hard M-A phases. As shown in Fig. 10b, this bimodal distribution facilitates the formation of interconnected high-KAM bands along the M-A chains. These continuous brittle paths essentially “short-circuit” the crack-arresting capability of effective high-angle grain boundaries (HAGBs, $\alpha > 45^\circ$) [2122]. Although the ICHAZ possesses a greater density of HAGBs, the interconnected M-A constituents provide a low-energy fracture expressway that guides the crack tip along the brittle interfaces. This guidance mechanism allows the crack to bypass ductile grain interiors where it would otherwise be arrested or significantly deflected by HAGBs.

In contrast, the CGHAZ maintains a more uniform SF distribution and isolated KAM zones, despite its coarser grains. In this region, the absence of interconnected brittle chains forces potential cracks to

interact directly with martensite laths and HAGBs, increasing the probability of crack arrest or deflection. Consequently, the synergy of M-A connectivity and strain localization in the ICHAZ overrides the benefits of grain refinement, rendering it the most vulnerable path for rapid crack coalescence under low-temperature loading.

Furthermore, the efficacy of crack resistance is closely related to the density of effective high-angle grain boundaries. While the ICHAZ possesses a higher absolute density of grain boundaries, the failure path is dictated by the interconnected M-A chains and the high-strain regions identified in the KAM maps. Consequently, under low-temperature impact loading, the synergy of soft/hard phase connectivity leads to a strain concentration in the ICHAZ that overrides the benefits of grain refinement. When these M-A particles form continuous chains, they create ‘brittle bridges’ at the grain boundaries. Cracks propagate preferentially along the low-energy interfaces of M-A chains or initiate microcracks directly from M-A sites, thereby effectively bypassing HAGBs, rendering them the most vulnerable region for crack initiation and rapid propagation.

4.2. Characteristics of M-A Chains in the ICHAZ

To clarify the influence of the M-A morphology and distribution in the ICHAZ on the low-temperature toughness, the aspect ratio (AR), elemental enrichment, average chain length, and chain connectivity index C of the M-A constituents were measured to reveal their impact mechanism. To quantify the parameters, the M-A constituents were classified in the following manner: the constituents were designated ‘blocky’ for $AR < 2$, ‘lath-like’ for $AR > 3$, and ‘transitional’ for $2 \leq AR \leq 3$. In accordance with the definition presented in Fig. 10, M-A constituents were designated chained if other M-A particles were observed within a distance equivalent to three times the equivalent circle diameter (d) of a specific M-A grain. The chaining index C is defined as follows (Eq. (1)):

$$C = (1 - E/N) * \bar{AR} \quad (1)$$

In the equation, N signifies the total number of distinct M-A phase particles identified within the analysed field of view; E indicates the number of independent objects (isolated particles or interconnected particle chains) determined statistically after segmentation of the binarised image. The value $(1 - E/N)$ has been demonstrated to measure the degree of particle aggregation. When the microstructure shows complete dispersion ($E \approx N$), this value approaches 0. Conversely, when particles form continuous chain-like structures ($E \ll N$), this value increases towards 1. Additionally, because lamellar M-A constituents pose

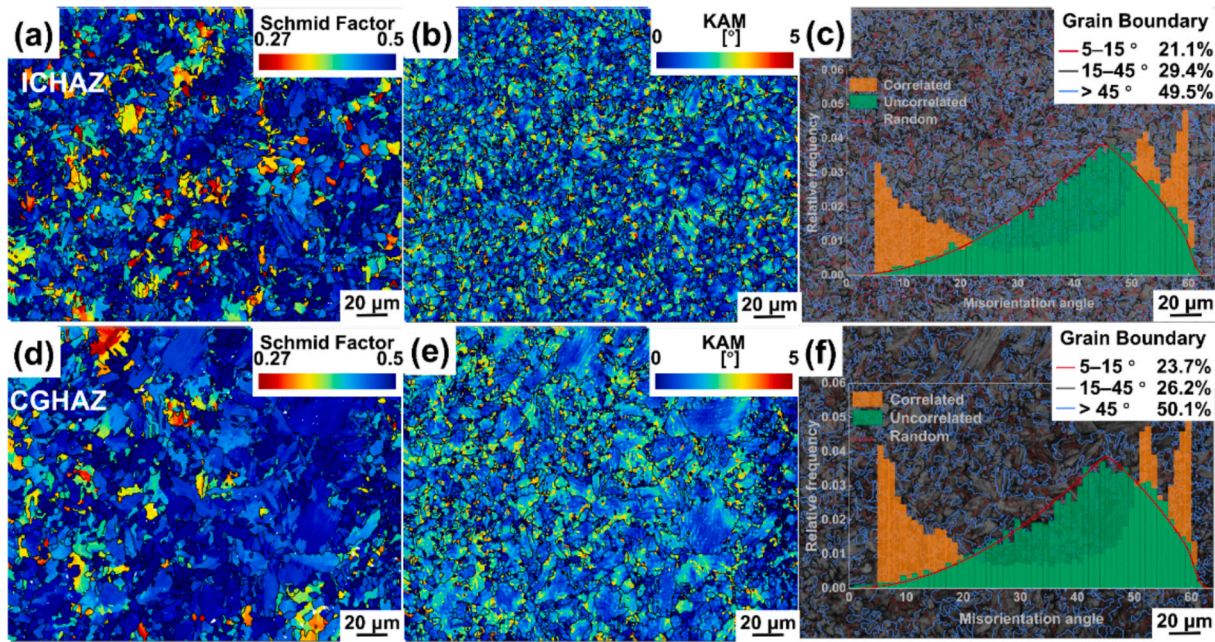


Fig. 10. EBSD analysis of the middle HAZ region. (a) Schmid factor of the ICHAZ; (b) KAM of the ICHAZ; (c) distribution of grain boundaries at different angles in the ICHAZ; (d) Schmid factor of the CGHAZ; (e) KAM of the CGHAZ; (f) distribution of grain boundaries at different angles in the CGHAZ.

a greater threat to material toughness and tend to align along grain boundaries, this model includes the term $\bar{A}R$ as a morphology weighting factor.

The M-A constituents are composed of high-carbon martensite and retained austenite. The brittleness of high-carbon martensite increases with increasing carbon content [23]. Figs. 11 and 12 show the C and Mn contents of the M-A constituents within the ICHAZ at 5.5 kW and 6.5 kW, respectively. A comparison of the carbon content of the base metal (0.06 wt.%) with that of the M-A constituents revealed a significant enrichment of carbon, especially in the samples from the middle region under different conditions. In contrast, manganese was less enriched; the maximum Mn content in M-A increased by only 0.26 wt.% compared with that in the base metal. The enrichment of Mn has been shown to increase the brittleness and hardness of M-A. At the same time, the ductile–brittle transition temperature increases [24–25], which amplifies the plasticity difference between M-A and the ferrite matrix.

As shown in Fig. 13, the distributions of the aspect ratios and the statistical morphology percentages of the M-A constituents in the ICHAZs of different regions under various laser powers are

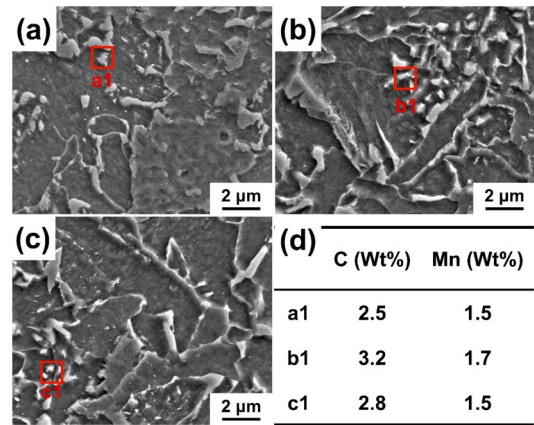


Fig. 12. SEM morphologies and C and Mn content analysis of M-A constituents in the ICHAZ at a laser power of 6.5 kW.

demonstrated. The statistical results indicate that the M-A constituents in the ICHAZ predominantly exist in blocky form. However, in the middle region of the sample, although blocky M-A remains dominant, its proportion is lower than that at other locations. Preliminary studies have indicated that the presence of lath-like M-A has a substantial effect on impact toughness [26]. Contrary to expectations, the proportion of lath-like M-A in the ICHAZ at 5.5 kW was greater than that at 6.5 kW in the middle region, which exhibited poorer toughness. Consequently, the morphology of individual M-A constituents does not represent the primary factor contributing to the diminished low-temperature impact toughness of the ICHAZ.

Fig. 14 presents the quantitative analysis of the mean chain length and chaining index C of M-A chains across different locations and laser powers. The results indicate that, at a constant power, the middle region has the highest mean chain length and chaining index compared with the upper and lower regions. For example, at 5.5 kW, the mean chain length in the middle region reaches 2.30 μm , and C reaches 1.33, which is greater than those in both the upper and lower regions, which reach 1.85 μm and 1.42 μm , respectively. Furthermore, increasing the laser

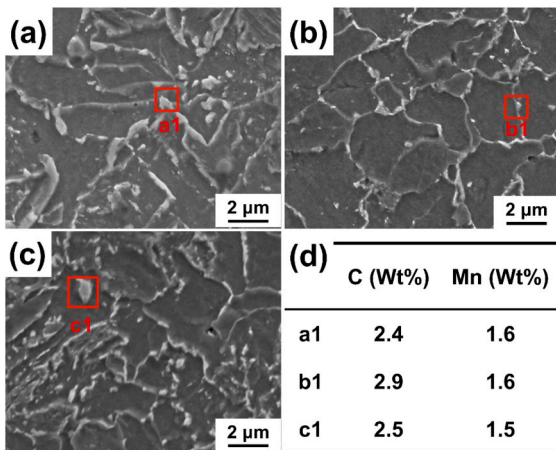


Fig. 11. SEM morphologies and C and Mn content analysis of M-A constituents in the ICHAZ at a laser power of 5.5 kW.

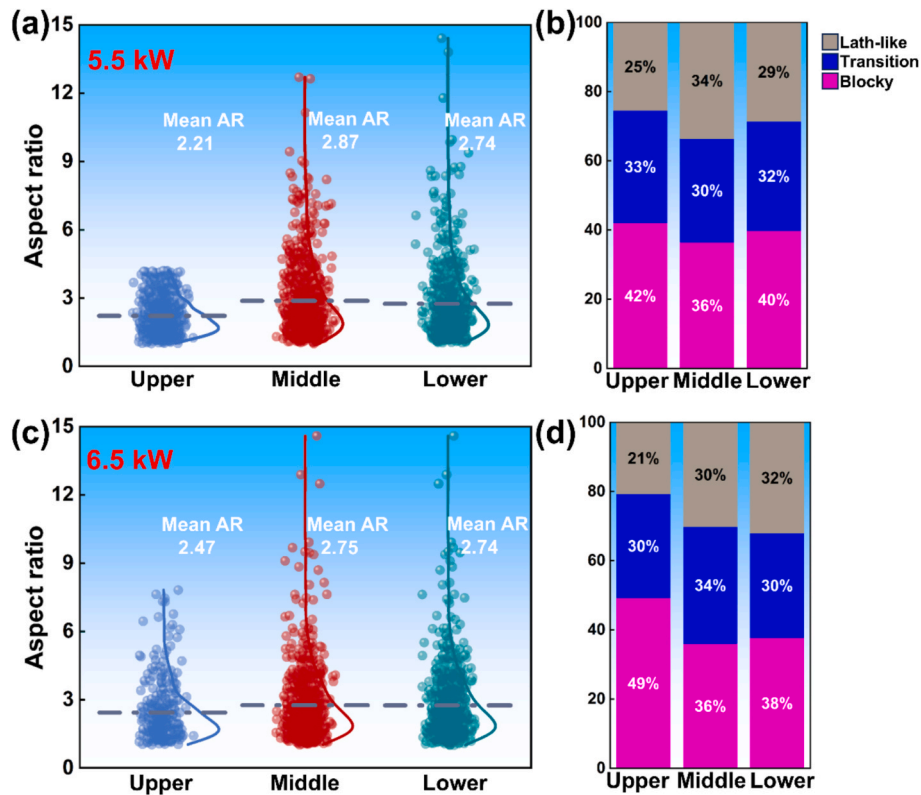


Fig. 13. Statistical characteristics of M-A constituents in the ICHAZ across the upper, middle, and lower regions: (a) aspect ratio (AR) distribution at 5.5 kW; (b) proportions of blocky ($AR < 3$), transitional ($3 \leq AR < 5$), and lath-like ($AR \geq 5$) morphologies at 5.5 kW; (c) aspect ratio distribution at 6.5 kW; (d) morphological proportions at 6.5 kW.

power to 6.5 kW leads to a significant increase in both metrics; specifically, in the middle region, the mean chain length increases to 2.74 μm , with a corresponding chaining index of 1.55. As visually confirmed in the skeletonized images, as shown in Fig. 14 b1, d1, and f1, the M-A chains at 6.5 kW not only display elongated main chains (red lines) but also substantially extended branch chains (white lines) compared with those under the 5.5 kW condition. These observations suggest that higher heat input enhances the propensity of M-A constituents to interconnect, leading to the formation of continuous brittle pathways characterized by increased chain lengths.

4.3. Mechanisms Description

Synthesizing the experimental findings reveals a distinct divergence in the embrittlement mechanisms of the HAZ subzones. In the CGHAZ, the M-A constituents predominantly exhibit a lath-like morphology and are distributed intragranularly along martensite laths. Despite the significant grain coarsening in this region, the high hardness of the matrix restricts plastic deformation under impact loading, effectively reducing localized stress accumulation. Furthermore, the high-angle boundaries of the martensite packets serve as effective barriers to crack propagation, thereby partially mitigating the detrimental effects of the brittle phase. Conversely, the fracture behavior in the ICHAZ is governed by the “chaining” of M-A constituents. This connectivity of brittle phases proves to be more deleterious than the grain coarsening observed in the CGHAZ. Consequently, the interconnected M-A chains render the fine-grained ICHAZ more susceptible to fracture than the coarse-grained region does, establishing M-A connectivity as the primary cause of the inferior toughness observed.

Research indicates that M-A chains significantly compromise the low-temperature toughness of the HAZ, acting as preferential sites for crack initiation. The fracture behaviour in the ICHAZ is governed by the

synergistic effect of the flow mismatch between M-A/ferrite and the chain-like distribution of M-A. Consistent with classical fracture mechanics theories [27,28], the connectivity of M-A constituents is equivalent to effectively increasing the size of the brittle phase. Although chaining may reduce the number density of independent particles, the detrimental effect of the increased equivalent effective particle size dominates the fracture process. Moreover, the stress fields generated by neighboring particles in the m-a chain combine, exacerbating the adverse effects on toughness. Consequently, these extended M-A networks facilitate crack propagation, rendering the region highly susceptible to brittle fracture. This finding supports the proposed “M-A chain-guided fracture mechanism”.

The process initiates with stress concentration and void nucleation, stages I and II, as shown in Fig. 15. Owing to the mechanical heterogeneity within the ICHAZ, a significant flow stress mismatch develops during plastic deformation between the hard, brittle M-A constituents and the softer surrounding ferrite. This generates intense stress concentrations, particularly at the sharp tips of lath-like M-A constituents, as shown in Fig. 15, in the first stage. Consequently, voids nucleate preferentially at these interfaces via decohesion [29]. While blocky M-A lacks sharp stress concentrators and initially remains inactive, lath-like M-A acts as the primary driver of damage initiation.

In the third stage, the subsequent stage involves void coalescence and microcrack linkage. With continued deformation, the stress field expands to involve blocky M-A, prompting secondary void formation. Crucially, the high chaining index and extended mean chain length—as quantitatively evidenced in the middle region statistics (Fig. 14)—greatly increase the probability of intervoid bridging. Voids at the tips of lath-like M-A extend parallel to the stress direction or merge with neighboring voids. Here, microcracks originating from blocky M-A function as “connectors” between chain segments, significantly shortening the coalescence distance. This linkage transforms isolated damage

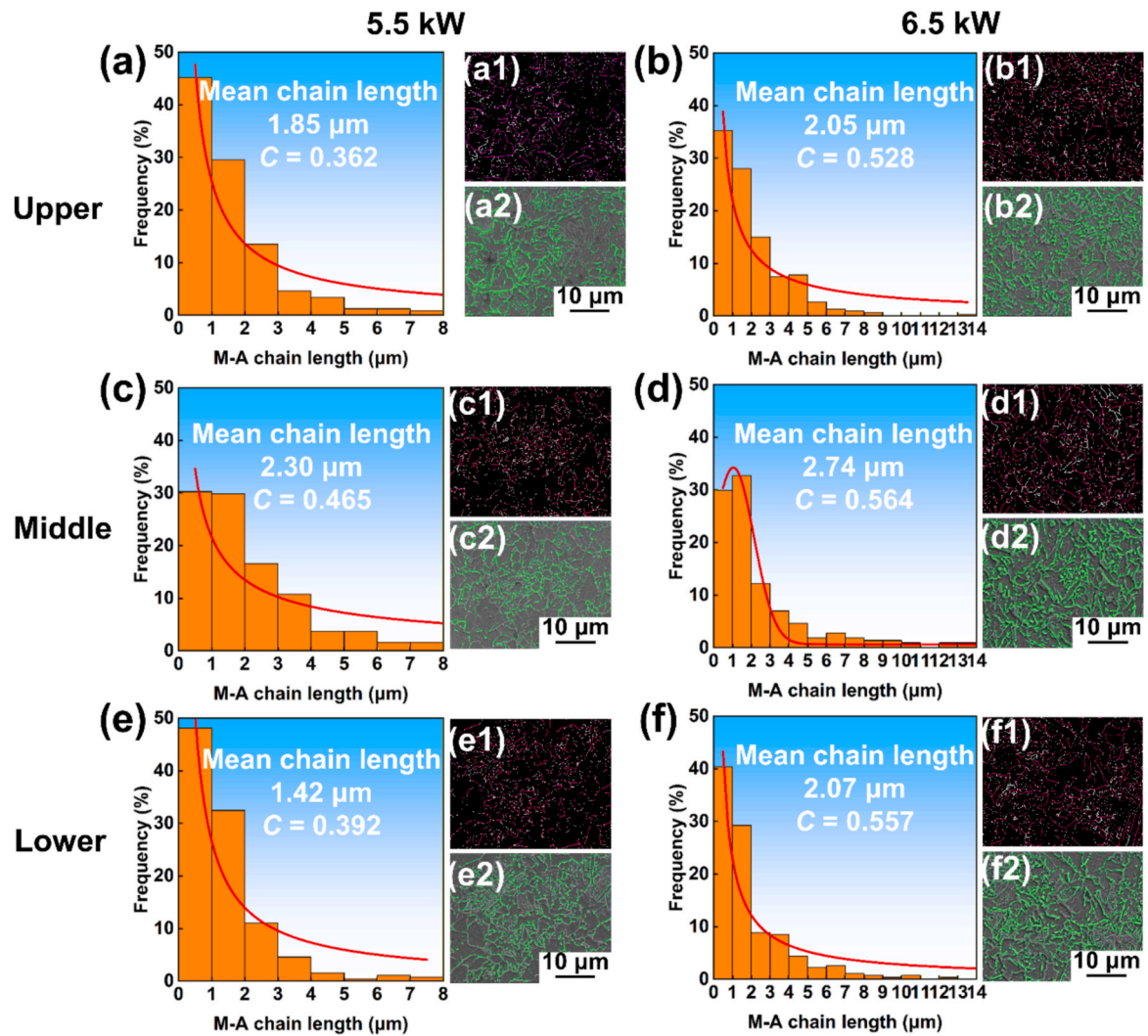


Fig. 14. Quantitative analysis of M-A chain connectivity in the ICHAZ. (a, c, e) Statistical distribution of M-A chain lengths in the upper, middle, and lower regions at 5.5 kW; (b, d, f) Statistical distribution of M-A chain lengths in the upper, middle, and lower regions at 6.5 kW; (a1 - e1) Schematic diagram of M-A chain skeletonization; the red line is the main chain, and the white line is the branch chain.

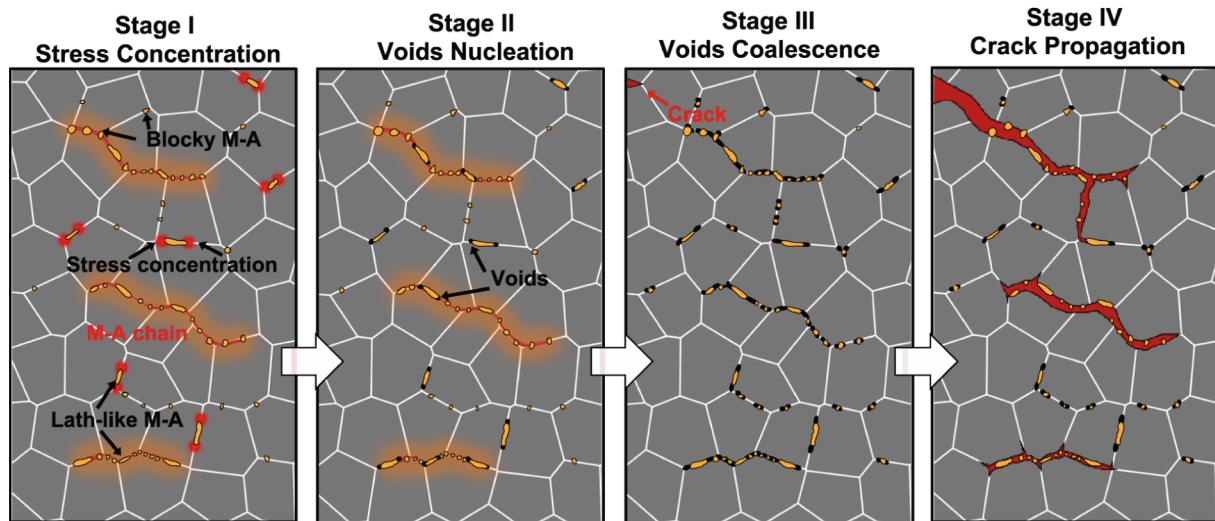


Fig. 15. Schematic diagram of the evolution process of crack initiation, propagation, and coalescence induced by M-A chains in the ICHAZ.

into a continuous, submacroscopic crack network.

The final stage is macroscopic crack propagation, Stage IV in Fig. 15.

Once the interconnected microcrack chain reaches a critical length, it establishes a low-energy pathway for instability. This enables the rapid, unstable propagation of the main crack along the M-A chain network, leading to definitive macroscopic failure.

In summary, the analysis demonstrates that the significant reduction in low-temperature toughness in the heat-affected zone caused by M-A chains is indeed evident, mainly due to their chain-like distribution characteristics, which is a pattern that encourages crack initiation and propagation. While this finding does not dismiss the potential influence of external variables, such as grain size, experimental evidence has confirmed the dominant role of the M-A chain-guided fracture mechanism in this process.

5. Conclusion

The experimental campaign performed in this study aimed to analyse the effects of laser power on the low-temperature impact toughness and corresponding microstructural evolution in the heat-affected zone of narrow-gap laser wire-feed welded EH36 steel. A multitechnique characterization approach was employed, combining impact testing with advanced electron microscopy and EBSD analysis, to establish the mechanistic link between process parameters and fracture behavior. The main conclusions of this paper can be summarized as follows:

- (1) The laser power exerts a significant and non-monotonic influence on the impact toughness of the HAZ, with absorbed energy peaking at approximately 240 J under 5.5 kW laser power. This represents an increase of over 60% compared to the 3.5 kW joint, followed by a marked deterioration at 6.5 kW, the result decreasing to 176 J. This toughness loss is accompanied by a distinct transition in fracture morphology from dimpled ductile to quasicleavage brittle, with the middle weld region consistently exhibiting the most severe embrittlement.
- (2) Microstructural analysis reveals that the chain-like distribution of M-A constituents in the ICHAZ is the dominant microstructural feature responsible for toughness degradation. With increasing laser power, both the average chain length and chaining index of M-A increase substantially. Specifically, in the middle region, the mean M-A chain length extended from 2.30 μm under 5.5 kW to 2.74 μm under 6.5 kW, while the chaining index *C* increased from 0.465 to 0.564. These expanded chains form continuous brittle pathways that intensify microstructural heterogeneity and promote microcrack nucleation and coalescence.
- (3) The fundamental mechanism of toughness reduction is governed by the M-A chain-guided fracture process, where the elastoplastic mismatch between M-A and the ferrite matrix causes localized stress concentration. Voids preferentially nucleate at lath-like M-A ends and chain intersections, propagating rapidly along paths of high strain localization. This mechanism establishes that the connectivity of brittle phases, rather than the grain size alone, controls the fracture path and toughness response at elevated power levels.

CRediT authorship contribution statement

Haiquan Guo: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Zhonglin Hou:** Writing – review & editing, Funding acquisition. **Baoping Wu:** Writing – review & editing, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

- [1] E.H. Cramer, R. Løseth, K. Olaisen, Fatigue assessment of ship structures, *Mar. Struct.* 8 (1995) 359–383, [https://doi.org/10.1016/0951-8339\(94\)00026-O](https://doi.org/10.1016/0951-8339(94)00026-O).
- [2] K. Guo, Z. Shi, T. Pan, X. Luo, F. Chai, Structure and morphology of M-A constituent in the CGHAZ of a quenched and tempered ultra-low-carbon steel, *J. Mater. Sci.* 58 (2023) 4587–4602, <https://doi.org/10.1007/s10853-023-08229-5>.
- [3] X. Yang, X. Di, X. Liu, D. Wang, C. Li, Effects of heat input on microstructure and fracture toughness of simulated coarse-grained heat affected zone for HSLA steels, *Mater. Charact.* 155 (2019) 109818, <https://doi.org/10.1016/j.matchar.2019.109818>.
- [4] Y. Li, S. Geng, J. Li, Z. Fan, C. Han, J. Jin, P. Jiang, Determination of residual stress in hybrid laser-arc welded U-rib-to-deck joints by thermo-metallurgical-mechanical simulation and neutron diffraction, *Thin-Walled Struct.* 208 (2025) 112755, <https://doi.org/10.1016/j.tws.2024.112755>.
- [5] A comparative study of laser-arc double-sided welding and double-sided arc welding of 6 mm 5A06 aluminium alloy, *Materials & Design.* 32 (2011) 2165–2171, <https://doi.org/10.1016/j.matdes.2010.11.041>.
- [6] M.M. Atabaki, N. Yazdian, J. Ma, R. Kovacevic, High power laser welding of thick steel plates in a horizontal butt joint configuration, *Opt. Laser Technol.* 83 (2016) 1–12, <https://doi.org/10.1016/j.optlastec.2016.03.016>.
- [7] S. Katayama, Y. Abe, R. Ido, M. Mizutani, Y. Kawahito, Deep penetration welding with high power disk lasers in low vacuum, in: *AIP Publishing* (2011) 669–678, <https://doi.org/10.2351/1.5062308>.
- [8] Z. Lei, H. Cao, X. Cui, Y. Ma, L. Li, Q. Zhang, A novel high efficiency narrow-gap laser welding technology of 120 mm high-strength steel, *Opt. Lasers Eng.* 178 (2024) 108232, <https://doi.org/10.1016/j.optlaseng.2024.108232>.
- [9] Z.-L. Hou, H.-Q. Guo, J.-J. Wang, Z.-Y. Huang, Z.-A. Wang, D.-S. Fang, J. Qiao, Research on the Low-Temperature Impact Toughness of a New 100-mm Ultra-Thick Offshore Steel Fabricated Using the Narrow-Gap Laser Wire Filling Welding Process, *Materials.* 17 (2024) 1363, <https://doi.org/10.3390/ma17061363>.
- [10] Z. Zhu, L. Min, B. Xiao, L. Xue, S. Zhao, J. Shen, C. Xu, Effect of Heat Input on the Metal Microstructure and Mechanical Properties of Q960 High-Strength Steel Solid Core Welding Wire, *Steel Res. Int.* 95 (2024) 2400060, <https://doi.org/10.1002/srin.202400060>.
- [11] S. Jia, Q. Ma, Y. Hou, B. Li, H. Zhang, Q. Liu, Changes in microstructure and properties of weld heat-affected zone of high-strength low-alloy steel, *J. Iron Steel Res. Int.* 31 (2024) 2041–2052, <https://doi.org/10.1007/s42243-023-01133-x>.
- [12] K.S. Arora, S.R. Pandu, N. Shajan, P. Pathak, M. Shome, Microstructure and impact toughness of reheated coarse grain heat affected zones of API X65 and API X80 linepipe steels, *Int. J. Press. Vessel. Pip.* 163 (2018) 36–44, <https://doi.org/10.1016/j.ijpvp.2018.04.004>.
- [13] X.L. Wang, Y.T. Tsai, J.R. Yang, Z.Q. Wang, X.C. Li, C.J. Shang, R.D.K. Misra, Effect of interpass temperature on the microstructure and mechanical properties of multi-pass weld metal in a 550-MPa-grade offshore engineering steel, *Weld, WORLD* 61 (2017) 1155–1168, <https://doi.org/10.1007/s40194-017-0498-x>.
- [14] W. Guo, L. Li, S. Dong, D. Crowther, A. Thompson, Comparison of microstructure and mechanical properties of ultra-narrow gap laser and gas-metal-arc welded S960 high strength steel, *Opt. Lasers Eng.* 91 (2017) 1–15, <https://doi.org/10.1016/j.optlaseng.2016.11.011>.
- [15] M. Zubairuddin, S.K. Albert, M. Vasudevan, S. Mahadevan, V. Chaudhari, V.K. Suri, Numerical simulation of multi-pass GTA welding of grade 91 steel, *J. Manuf. Process.* 27 (2017) 87–97, <https://doi.org/10.1016/j.jmapro.2017.04.031>.
- [16] Z.F. Zhang, D.M. Tong, X.Y. Yang, X.F. Wang, L.Z. Han, G.H. Yan, C.W. Li, J.F. Gu, In-situ research on tensile deformation and microvoid formation in a nuclear pressure vessel steel, *Int. J. Miner. Metall. Mater.*, 32(2025), No. 8, pp.1893-1901, <https://doi.org/10.1007/s12613-024-3043-0>.
- [17] N.Y. Du, H.H. Liu, Y.F. Cao, P.X. Fu, C. Sun, H.W. Liu, D.Z. Li, In situ investigation of the fracture of primary carbides and its mechanism in M50 steel, *Mater. Charact.* 186 (2022) 111822, <https://doi.org/10.1016/j.matchar.2022.111822>.
- [18] N.A. Giang, M. Kuna, G. Hütter, Influence of carbide particles on crack initiation and propagation with competing ductile-brittle transition in ferritic steel, *Theor. Appl. Fract. Mech.* 92 (2017) 89–98, <https://doi.org/10.1016/j.tafmec.2017.05.015>.
- [19] Y. Terada, H. Ishikawa, R. Chijiwa, M. Sugiyama, H. Tamehiro, Sour-Resistant X65 UOE Line Pipe for Low-Temperature Service, *Int. J. Offshore Polar Eng.* 20 (2010) 218–223.
- [20] F.F. Giginyak, Cottrell-Petch criterion of brittle fracture, *Strength Mater.* 12 (1980) 304–307, <https://doi.org/10.1007/BF01129078>.
- [21] Effect of cooling rate on intercritically reheated microstructure and toughness in high strength low alloy steel: *Materials Science and Technology: Vol 9, No 1, (n.d.)*, <https://www.tandfonline.com/doi/abs/10.1179/mst.1993.9.1.8> (accessed September 29, 2025).

- [22] A.-F. Gourgues, H.M. Flower, T.C. Lindley, Electron backscattering diffraction study of acicular ferrite, bainite, and martensite steel microstructures, *Mater. Sci. Technol.* 16 (2000) 26–40, <https://doi.org/10.1179/026708300773002636>.
- [23] L. Duan, Y. Chen, Q. Liu, S. Jia, C. Jia, Microstructures and Mechanical Properties of X100 Pipeline Steel Strip, *J. Iron Steel Res. Int.* 21 (2014) 227–232, [https://doi.org/10.1016/S1006-706X\(14\)60035-2](https://doi.org/10.1016/S1006-706X(14)60035-2).
- [24] J. Li, X. Xiong, Q. Shen, Q. Zhai, W. Liu, Effect of Mn on precipitation behaviour of NiAl phase in Fe–NiAl steels, *Mater. Sci. Technol.* 36 (2020) 852–857, <https://doi.org/10.1080/02670836.2020.1745410>.
- [25] Y. Yang, X. Pan, Effect of Mn/N ratio on microstructure and mechanical behavior of simulated welding heat affected zone in 22% Cr lean duplex stainless steel, *Mater. Sci. Eng. A* 835 (2022) 142676, <https://doi.org/10.1016/j.msea.2022.142676>.
- [26] Z. Wang, M. Shi, S. Tang, G. Wang, Effect of heat input and M-A constituent on microstructure evolution and mechanical properties of heat affected zone in low carbon steel, *J. Wuhan Univ. Technol.-Mat. Sci. Edit.* 32 (2017) 1163–1170, <https://doi.org/10.1007/s11595-017-1726-3>.
- [27] D.A. Curry, J.F. Knott, Effect of microstructure on cleavage fracture toughness of quenched and tempered steels, *Metal, Science* 13 (1979) 341–345, <https://doi.org/10.1179/msc.1979.13.6.341>.
- [28] Effect of carbide distribution on the fracture toughness in the transition temperature region of an SA 508 steel, *Acta Materialia*. 50 (2002) 4755–4762. [https://doi.org/10.1016/S1359-6454\(02\)00313-0](https://doi.org/10.1016/S1359-6454(02)00313-0).
- [29] Rupture mechanisms in combined tension and shear—Micromechanics, *International Journal of Solids and Structures*. 44 (2007) 5481–5498. <https://doi.org/10.1016/j.ijsolstr.2007.01.010>.